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Changing APN functions at two points

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Abstract
We investigate a construction in which a vectorial Boolean function G is obtained from a given function F over $\mathbb{F}_{2^n}$ by changing the values of F at two points of the underlying field. In particular, we examine the possibility of obtaining one APN function from another in this way. We characterize the APN-ness of G in terms of the derivatives and in terms of the Walsh coefficients of F. We establish that changing two points of a function F over $\mathbb{F}_{2^n}$ which is plateaued (and, in particular, AB) or of algebraic degree $deg(F) < n - 1$ can never give a plateaued (and AB, in particular) function for any $n \geq 5$. We also examine a particular case in which we swap the values of F at two points of $\mathbb{F}_{2^n}$. This is motivated by the fact that such a construction allows us to obtain one permutation from another. We obtain a necessary and sufficient condition for the APN-ness of G which we then use to show that swapping two points of any power function over a field $\mathbb{F}_{2^n}$ with $n \geq 5$ can never produce an APN function. We also list some experimental results indicating that the same is true for the switching classes from Edel and Pott (Adv. Math. Commun. **3**(1):59–81, 2009), and conjecture that the Hamming distance between two APN functions cannot be equal to two for $n \geq 5$.

Keywords Almost perfect nonlinear functions · Differential uniformity · Boolean functions

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1 Introduction

A construction in which a given vectorial Boolean function is modified at one point are investigated in [2] in the context of the problem of the maximum algebraic degree of an APN function: given a function $F : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$ and $u, v \in \mathbb{F}_{2^n}$, the one-point modification of F is defined as $G(x) = F(x) + (1 + (x + u)^{2^n-1})v$ so that $G(u) = F(u) + v$ and $G(x) = F(x)$ for $x \neq u$; the reasoning behind this is that any G with algebraic degree

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$\deg(G) = n$ can be expressed in this form. Although the question of the existence of an APN function of algebraic degree n over $\mathbb{F}_{2^n}$ remains open in general, a multitude of non-existence results are derived in [2] which support the conjecture that this is impossible. In this paper we examine a similar construction involving two points. More precisely, given $F : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$, we select $u_1, u_2 \in \mathbb{F}_{2^n}^*$ and $v_1, v_2 \in \mathbb{F}_{2^n}$, and define $G(x) = F(x) + (1 + (x+u_1)^{2^n-1})v_1 + (1+(x+u_2)^{2^n-1})v_2$. We examine under what conditions G can be APN, especially when F itself is APN. We see that if F is plateaued or has $\deg(F) < n-1$, then G is not plateaued for $n \geq 5$; in particular, G is not AB if F is AB for $n \geq 5$. We also observe that if F is quadratic and not APN, then changing two points of F cannot produce an APN function either.

We also examine a special case in which the values of F at u_1 and u_2 are swapped, so that $G(u_1) = F(u_2)$ and $G(u_2) = F(u_1)$; under EA-equivalence, we also assume $u_1 = 0$ and $u_2 = 1$. This case is more tractable and is interesting in that it allows us to obtain one permutation from another. We characterize the APN-ness of $G(x) = F(x) + x^{2^n-1} + (x+1)^{2^n-1}$ in terms of the values $\sum_{y\in\mathbb{F}_{2^n}} \Delta_F(y, F(y)+1)$ and $\sum_{y\in\mathbb{F}_{2^n}} \Delta_F(y+1, F(y))$, where $\Delta_F(a,b)$ is the number of solutions x to $D_aF(x) = b$. We demonstrate how a formula for the first of these values can be derived for $F(x) = x^3$. More generally, we show that no APN function can be obtained by swapping the values at $u_1 = 0$ and $u_2 = 1$ of any power function over $\mathbb{F}_{2^n}$ for $n \geq 5$, and verify that this is also true for all APN functions from [10]. Based on these results, we conjecture that swapping two points in an arbitrary APN function can never produce an APN function for $n \geq 5$.

The method of swapping two points has previously been investigated in the context of constructing 4-differentially uniform permutations from the inverse function [16]. Some basic properties are given in [16], and the authors generalize the construction by changing the values of points lying on a cycle of length greater than two [13]; in particular, they obtain some involutions using the method (see [9]).

2 Preliminaries

2.1 Representation of vectorial functions

An *n-dimensional Boolean function* is any function $f : \mathbb{F}_{2^n} \to \mathbb{F}_2$. A *vectorial Boolean* (n,m)*-function* is any function $F : \mathbb{F}_2^n \to \mathbb{F}_2^m$. We focus only on the case $m = n$, in which it is more convenient to identify the vector space $\mathbb{F}_2^n$ with the finite field $\mathbb{F}_{2^n}$ and to consider functions from $\mathbb{F}_{2^n}$ to itself. Any (n,n)-function has a unique representation as a univariate polynomial over $\mathbb{F}_{2^n}$ of the form

$$F(x) = \sum_{i=0}^{2^n-1} a_i x^i, a_i \in \mathbb{F}_{2^n}.$$

Given a natural number $0 \leq i \leq 2^n - 1$ and its binary expansion $i = \sum_{k=0}^{n-1} 2^k i_k$ with $i_k \in \{0,1\}$, its *two-weight* $w_2(i)$ is the number of nonzero terms in its binary expansion, i.e. $w_2(i) = \sum_{k=0}^{n-1} i_k$. The *algebraic degree* of F is then

$$\deg(F) = \max\{w_2(i) : 0 \leq i \leq 2^n - 1, a_i \neq 0\}.$$

Given an (n,n)-function F, its *component functions* are the functions from $\mathbb{F}_{2^n}$ to $\mathbb{F}_2$ of the form $\mathrm{Tr}_1^n(bF(x))$ for $b \in \mathbb{F}_{2^n}$, where Tr_1^n is the absolute trace function from $\mathbb{F}_{2^n}$ to $\mathbb{F}_2$, which we denote simply by Tr if the dimension n is clear from the context.

2.2 Almost perfect nonlinear functions and bent functions

Let F be an (n, n)-function. The *derivative of F in direction a* for any $a \in \mathbb{F}_{2^n}$ is the function $D_aF : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$ defined as $D_aF(x) = F(x) + F(a + x)$. The *differential sets* H_aF are the image sets of the derivatives of F, i.e. the sets of the form $H_aF = \{D_aF(x) : x \in \mathbb{F}_{2^n}\} = \{F(x) + F(a + x) : x \in \mathbb{F}_{2^n}\}$.

For convenience, we also define the *shifted derivative* $D_a^\beta F$ in direction $a \in \mathbb{F}_{2^n}$ with shift $\beta \in \mathbb{F}_{2^n}$ as the function $D_a^\beta F(x) = F(x) + F(a + x) + F(a + \beta)$ and the *shifted differential set* $H_a^\beta F$ in direction a with shift β as its image set, i.e. as $H_a^\beta F = \{D_a^\beta F(x) : x \in \mathbb{F}_{2^n}\} = \{F(x) + F(a + x) + F(a + \beta) : x \in \mathbb{F}_{2^n}\}$.

For any $a, b \in \mathbb{F}_{2^n}$, define $\Delta_F(a, b) = |\{x \in \mathbb{F}_{2^n} : F(x + a) + F(x) = b\}|$. Then, the *differential uniformity* of F is defined as $\Delta_F = \max\{\Delta_F(a, b) : a, b \in \mathbb{F}_{2^n}, a \neq 0\}$. An (n, n)-function F is called *differentially δ-uniform* if $\Delta_F \leq \delta$. If $\delta = 2$, then F is called *almost perfect nonlinear* (APN). Note that $\Delta_F \geq 2$ for any (n, n)-function and hence APN functions are those with optimal differential uniformity.

APN functions over $\mathbb{F}_{2^n}$ can be characterized in several different ways. In this paper we focus on characterizations by means of the differential properties and the power moments of the function's Walsh transform. The *Walsh transform* of a Boolean function $f : \mathbb{F}_{2^n} \to \mathbb{F}_2$ is defined by

$$W_f(a) = \sum_{x \in \mathbb{F}_{2^n}} (-1)^{f(x)+\mathrm{Tr}_1^n(ax)}, \quad a \in \mathbb{F}_2$$

for any $a \in \mathbb{F}_{2^n}$. Also useful is the inverse Walsh transform formula

$$\sum_{a \in \mathbb{F}_{2^n}} W_f(a) = 2^n (-1)^{f(0)}. \tag{1}$$

The *Walsh transform* of an (n, n)-function F is defined by

$$W_F(a, b) = \sum_{x \in \mathbb{F}_{2^n}} (-1)^{\mathrm{Tr}_1^n(bF(x))+\mathrm{Tr}_1^n(ax)} \tag{2}$$

for any $a, b \in \mathbb{F}_{2^n}$.

The Boolean functions $\mathrm{Tr}_1^n(bF(x))$ for $b \in \mathbb{F}_{2^n}^*$ are said to be the *component functions of F*. In this way, the values of the Walsh transform of an (n, n)-function can be seen as the values of the Walsh transform of the component functions of F.

In order to simplify notation, we will use $\eta(X)$, resp. $\eta_b(X)$ as shorthand for $(-1)^{\mathrm{Tr}_1^n(X)}$, resp. $(-1)^{\mathrm{Tr}_1^n(bX)}$ where X is an arbitrary expression and $b \in \mathbb{F}_{2^n}$. We also define the *equality indicator* $I(A, B)$, where A and B are some arbitrary expressions, as $I(A, B) = 1$ if $A = B$ and $I(A, B) = 0$ otherwise.

Given a set S, its characteristic function is denoted by $1_S(x)$ so that we have $1_S(x) = 1$ for $x \in S$ and $1_S(x) = 0$ for $x \notin S$. For a finite set $S = \{s_1, s_2, \ldots, s_k\}$ we write $1_{s_1,s_2,\ldots,s_k}(x)$ as shorthand for $1_{\{s_1,s_2,\ldots,s_k\}}(x)$.

With the above background, we can give the following classic characterizations of APN functions in terms of their Walsh transform.

Lemma 1 (see e.g. [8]) *Let F be an (n, n)-function. Then F is APN if and only if*

$$\sum_{a \in \mathbb{F}_{2^n}} \sum_{b \in \mathbb{F}_{2^n}^*} W_F^4(a, b) = 2^{3n+1}(2^n - 1).$$

Lemma 2 ([4, 5]) *Let F be an APN function over $\mathbb{F}_{2^n}$ satisfying $F(0) = 0$. Then*

$$\sum_{a,b\in\mathbb{F}_{2^n}} W_F^3(a,b) = 3\cdot 2^{3n} - 2^{2n+1}.$$

If F is plateaued , this condition is necessary and sufficient for F to be APN (plateaued functions are defined in Section 2.3 *below).*

The nonlinearity N_F of an (n,n)-function F is the minimum Hamming distance between its component functions and the affine functions. It measures the function's resistance to linear cryptanalysis [14] and equals

$$N_F = 2^{n-1} - \frac{1}{2}\max_{a\in\mathbb{F}_{2^n}, b\in\mathbb{F}_{2^n}^*} |W_F(a,b)|. \tag{3}$$

An (n,m)-function F is called *bent* if $W_F(a,b) \in \{\pm 2^{n/2}\}$ for all $a \in \mathbb{F}_{2^n}$ and nonzero $b \in \mathbb{F}_{2^m}^*$. Clearly, bent functions can exist only for even values of n in which case they achieve optimum nonlinearity. Nyberg in [15] proved that (n,m)-bent functions exist if and only if n is even and $m \leq n/2$. When n is odd or $n = m$, the optimal functions with respect to nonlinearity are the almost bent functions. An (n,n)-function F is called *almost bent* (AB) if its Walsh transform satisfies $W_F(a,b) \in \{0, \pm 2^{(n+1)/2}\}$ for any $a \in \mathbb{F}_{2^n}$ and nonzero $b \in \mathbb{F}_{2^n}$. Any AB function is APN, but not vice versa. However, for n odd, every quadratic APN function is also AB [6]; more generally, every plateaued APN function is also AB.

2.3 Plateaued functions

A Boolean function $f : \mathbb{F}_{2^n} \to \mathbb{F}_2$ is called *plateaued* if its Walsh transform satisfies $W_f(a) \in \{0, \pm\mu\}$ for some positive integer μ called the *amplitude* of f. An (n,n)-function F is called *plateaued* if all of its component functions are plateaued and it is called *plateaued with single amplitude* if all of its component functions are plateaued and have the same amplitude.

The amplitude of a plateaued Boolean function f is always of the form 2^λ for some $\lambda \geq \frac{n}{2}$, due to the well-known *Parseval's identity* $\sum_{a\in\mathbb{F}_{2^n}} W_f^2(a) = 2^{2n}$.

Since the algebraic degree of a Boolean plateaued function in n variables with amplitude μ is upper bounded by $n - \lambda + 1$ [12], the algebraic degree of a plateaued (n,n)-function F is upper bounded by $\max_{b\in\mathbb{F}_{2^m}^*}(n - \lambda_b + 1)$ where λ_b is the amplitude of $\mathrm{Tr}_1^m(bF(x))$, $b \neq 0$, so that $W_F(a,b) \in \{0, \pm 2^{\lambda_u}\}$ for any $a \in \mathbb{F}_{2^n}$. Since there exists no bent (n,n)-function, this maximum is less than or equal to $n - (n+1)/2 + 1 = (n+1)/2$. Hence a plateaued function can have algebraic degree n, resp. $n-1$ only if $n \leq 1$, resp. $n \leq 3$.

2.4 Equivalence relations of functions

There are several equivalence relations of functions for which differential uniformity and nonlinearity are invariant. Due to these equivalence relations, having only one APN (respectively, AB) function, one can generate a huge class of APN (respectively, AB) functions.

Two (n,n)-functions F and F' are said to be

- *affine equivalent* (*linear equivalent*) if $F' = A_1 \circ F \circ A_2$, where A_1 and A_2 are affine (linear) permutations of $\mathbb{F}_{2^n}$;

- *extended affine equivalent* (EA-equivalent) if $F' = A_1 \circ F \circ A_2 + A$, where A, A_1, A_2 are affine mappings over $\mathbb{F}_{2^n}$ and A_1, A_2 are permutations;
- *Carlet-Charpin-Zinoviev equivalent* (CCZ-equivalent) if for some affine permutation $\mathcal{L}$ of $\mathbb{F}_{2^n} \times \mathbb{F}_{2^n}$ the image of the graph of F is the graph of F', that is, $\mathcal{L}(G_F) = G_{F'}$ where $G_F = \{(x, F(x)) : x \in \mathbb{F}_{2^n}\}$ and $G_{F'} = \{(x, F'(x)) : x \in \mathbb{F}_{2^n}\}$.

It is obvious that linear equivalence is a particular case of affine equivalence, and that affine equivalence is a particular case of EA-equivalence. As shown in [6], EA-equivalence is a particular case of CCZ-equivalence and every permutation is CCZ-equivalent to its inverse. The algebraic degree of a function (if it is not affine) is invariant under EA-equivalence but, in general, it is not preserved by CCZ-equivalence. The following proposition illustrates how EA-equivalence can be expressed in terms of CCZ-equivalence.

Proposition 3 [3] *Let F and F' be functions from $\mathbb{F}_{2^n}$ to itself. The function F' is EA-equivalent to the function F or to the inverse of F (if it exists) if and only if there exists an affine permutation $\mathcal{L} = (L_1, L_2)$ on $\mathbb{F}_2^{2n}$ such that $\mathcal{L}(G_F) = G_{F'}$ and the function L_1 depends only on one variable, i.e. $L_1(x, y) = L(x)$ or $L_1(x, y) = L(y)$.*

It is worth mentioning some properties that remain invariant under CCZ-equivalence. Assuming that F and F' are CCZ-equivalent, we have:

- $\{\Delta_F(a, b) : a, b \in \mathbb{F}_{2^n}, a \neq 0\} = \{\Delta_{F'}(a, b) : a, b \in \mathbb{F}_{2^n}, a \neq 0\}$ [1];
- if F is APN then F' is APN too;
- $N_F = N_{F'}$ [6];
- if F is AB then F' is AB too;
- if F is plateaued with single amplitude λ then F' is plateaued with the same single amplitude λ;
- if F is plateaued with different amplitudes then F' is not necessarily plateaued: it can happen that F' has no plateaued components at all. However, if F and F' are EA-equivalent then F' is plateaued with the same multi-set of amplitudes.

3 Changing two points of a given function

The properties of a construction that involves changing precisely one value of given function F in order to obtain a new function G are investigated in [2]. The main point of interest is the possibility of obtaining an APN function G in this manner, and particular attention is paid to the case when the given F is itself APN. This is motivated by the open problem of the existence of APN functions of algebraic degree n over the field $\mathbb{F}_{2^n}$, since if two functions F and G are at distance one, then at least one of them must be of algebraic degree n. Two main characterizations of the APN-ness of such functions G are obtained in [2], one involving the Walsh coefficients of F, and one based on the properties of the derivatives $D_a F$ of F themselves. These characterizations are then applied in order to conclude that any G obtained by such a one-point change from a given F cannot be APN (except possibly for $n < 3$ in some cases) if F is a power, plateaued, quadratic or almost bent function . A number of additional non-existence results are also shown, which also agree with the conjecture that no APN function of algebraic degree n exists over $\mathbb{F}_{2^n}$.

This construction can naturally be generalized so as to encompass more than one point. Even for a small, fixed number of points $u_1, u_2, \ldots u_K$ at which the function is to be changed, however, the problem of the APN-ness of the resulting function seems very hard.

In the following we investigate whether, and under what conditions it is possible to obtain an APN function by changing the values of two distinct points in a given APN function F. More precisely, given two distinct elements u_1, u_2 from $\mathbb{F}_{2^n}$ and two arbitrary elements v_1, v_2 from $\mathbb{F}_{2^n}$, we are interested in the APN-ness of the function

$$\begin{aligned} G(x) &= F(x) + 1_{u_1}(x)v_1 + 1_{u_2}(x)v_2 = F(x) + (1 + (x + u_1))^{2^n-1}v_1 \\ &+(1 + (x + u_2))^{2^n-1}v_2 = \begin{cases} F(x) + v_1 & x = u_1 \\ F(x) + v_2 & x = u_2 \\ F(x) & x \notin \{u_1, u_2\}. \end{cases} \end{aligned} \tag{4}$$

We take $v_1 = v_2$ and denote both v_1 and v_2 by v since assuming otherwise leads to the problem of the existence of APN functions of algebraic degree n already discussed in [2]. We will also assume $v_1, v_2 \neq 0$ so that the distance between F and G is indeed equal to two.

A natural way to investigate the differential properties of G is to examine the derivatives D_aG and their expression via the derivatives D_aF of F. From the definition of G in (4) we can immediately see that for any $a \in \mathbb{F}_{2^n}^*$, the derivative D_aG takes the form

$$D_aG(x) = D_aF(x) + 1_{u_1,a+u_1}(x)v_1 + 1_{u_2,a+u_2}(x)v_2. \tag{5}$$

Recall that a function F over $\mathbb{F}_{2^n}$ is APN if and only if there do not exist elements $a \in \mathbb{F}_{2^n}^*$ and $x, y \in \mathbb{F}_{2^n}$ such that $D_aF(x) = D_aF(y)$ and $x + y \notin \{0, a\}$; this follows immediately from the definition. A characterization of the conditions under which G is APN can then be derived immediately from (5) by examining under what conditions such a triple of elements $(a, x, y) \in \mathbb{F}_{2^n}^3$ violating this condition may exist.

Proposition 4 *Let F be an (n, n)-function and let u_1, u_2 be two distinct points from $\mathbb{F}_{2^n}$. Let also v_1, v_2 be arbitrary elements from $\mathbb{F}_{2^n}$. Then the function*

$$G(x) = F(x) + 1_{u_1}(x)v_1 + 1_{u_2}(x)v_2$$

is APN if and only if all of the following conditions are satisfied for every derivative direction $a \in \mathbb{F}_{2^n}^*$:

(i) *D_aF is 2-to-1 on $\mathbb{F}_{2^n} \setminus \{u_1, u_2, a + u_1, a + u_2\}$;*
(ii) *$D_aF(u_1) + D_aF(u_2) \neq v_1 + v_2$ unless $a = u_1 + u_2$;*
(iii) *$D_aF(x) + D_aF(u_i) \neq v_i$ unless $a = u_1 + u_2$ or $x \in \{u_1, u_2, a + u_1, a + u_2\}$, for $i \in \{1, 2\}$;*
(iv) *$D_{u_1+u_2}F(u_i) + D_{u_1+u_2}F(x) \neq v_1 + v_2$ unless $x \in \{u_1, u_2\}$, for $i \in \{1, 2\}$.*

Proof From the definition, G is APN if and only if $D_aG(x) = D_aG(y)$ implies $a = 0$ or $x + y \in \{0, a\}$ for any $a, x, y \in \mathbb{F}_{2^n}$. We now examine under what circumstances such an equality may occur.

Suppose that $D_aG(x) = D_aG(y)$ for some $a, x, y \in \mathbb{F}_{2^n}$ with $a \neq 0$, $x \neq y$ and $x \neq a + y$. Depending on whether x and y are in $\{u_1, u_2, a + u_1, a + u_2\}$, we examine the following cases:

(i) If neither x nor y is in $\{u_1, u_2, a+u_1, a+u_2\}$, then $D_aG(x) = D_aG(y)$ is equivalent to $D_aF(x) = D_aF(y)$. However, due to the assumption $x \notin \{y, a + y\}$, this means that D_aF itself is not 2-to-1.
(ii) If say $x \in \{u_1, a + u_1\}$ and $y \in \{u_2, a + u_2\}$, then $D_aG(x) = D_aG(y)$ becomes $D_aF(u_1) + D_aF(u_2) = v_1 + v_2$ since $x \neq y + a$ implies that $a \neq u_1 + u_2$.
(iii) If say $x \in \{u_1, a + u_1\}$ but $y \notin \{u_1, u_2, a + u_1, a + u_2\}$ and $a \neq u_1 + u_2$, then $D_aG(x) = D_aG(y)$ is equivalent to $D_aF(u_1) + D_aF(y) = v_1$.

(iv) If $a = u_1 + u_2$ and say $x \in \{u_1, u_2\}$, $y \notin \{u_1, u_2\}$, then $D_a G(x) = D_a G(y)$ becomes $D_a F(u_1) + D_a F(y) = v_1 + v_2$.

Thus, all of the four conditions listed in the statement of the proposition are clearly necessary for G to be APN. However, they are also sufficient since if $D_a G(x) = D_a G(y)$ is true for some $a, x, y \in \mathbb{F}_{2^n}$, with $a \neq 0$ and $x + y \notin \{0, a\}$, then one of these four cases must necessarily occur. □

One can easily find functions F and elements u_1, u_2, v_1, v_2 satisfying the conditions of Proposition 4 in $\mathbb{F}_{2^n}$ with $n \leq 4$. For example, taking $F(x) = x^3$ over $\mathbb{F}_{2^4}$, $u_1 = 0$, $u_2 = \alpha^k$ and $v_1 = v_2 = \alpha^{3k}$ for some positive integer k, it is obvious that conditions (i) and (ii) of the proposition are immediately satisfied (on account of F being APN), and it can be checked computationally that conditions (iii) and (iv) are satisfied as well. Note that due to $v_1 = v_2$, this corresponds to the case of "swapping" two points described in Section 5. More generally, for any $u_1, u_2 \in \mathbb{F}_{2^4} \times \mathbb{F}_{2^4}$ with $u_1 \neq u_2$, there exists precisely one pair v_1, v_2 such that G from the statement of Proposition 4 is APN; furthermore, $v_1 = v_2$ in all such cases.

For dimensions $n \geq 5$, the question of whether an APN function can be obtained by changing (or, in particular, swapping) two of the values of a another APN function remains unsolved. We present some partial negative results in Sections 5 and 6, but the problem of the minimum distance between APN functions remains open.

Corollary 5 *Let F be a quadratic function over $\mathbb{F}_{2^n}$. If F is not APN then $G(x) = F(x) + 1_{u_1}(x)v_1 + 1_{u_2}(x)v_2$ is not APN either.*

Proof If F is quadratic, then all derivatives $D_a F$ for $a \neq 0$ are 2^{s_a}-to-1 functions, with $1 \leq s_a \leq n$. Furthermore, if F is not APN, then for some a we must have $s_a > 1$. Hence G is not APN by condition (i) in Proposition 4. □

4 Characterization by means of the Walsh transform

From (4) and assuming $v_1 = v_2 = v$, the Walsh coefficients of G can be expressed as $W_G(a, b) = W_F(a, b) + [\eta(bv) - 1][\eta(bF(u_1) + au_1) + \eta(bF(u_2) + au_2)]$. The difference of $W_F(a, b)$ and $W_G(a, b)$ can thus take only a limited number of different values:

$$W_G(a,b) - W_F(a,b) = \begin{cases} 4 & \mathrm{Tr}(bv) = \mathrm{Tr}(bF(u_1) + au_1) = \mathrm{Tr}(bF(u_2) + au_2) = 1 \\ -4 & \mathrm{Tr}(bv) = 1, \mathrm{Tr}(bF(u_1) + au_1) = \mathrm{Tr}(bF(u_2) + au_2) = 0 \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

In light of (3) this then implies that the nonlinearity of G can differ from that of F by at most two.

Recall that at least one of F and G must have algebraic degree equal to $(n - 1)$. Since the algebraic degree of AB functions is upper bounded by $\frac{n+1}{2}$ [6] then G is not AB when F is (for $n \geq 4$), and, therefore, $N_G = 2^{n-1} - 2^{\frac{n-1}{2}} - 2$. Moreover if F is plateaued then its algebraic degree is upper bounded by $\max_{b \in \mathbb{F}_{2^n}^*} (n - \lambda_b + 1)$ where 2^{λ_b} is the amplitude of $\mathrm{Tr}_1^m(bF(x))$, $b \neq 0$, and, therefore, G cannot be plateaued for $n \geq 5$. The upper bound on algebraic degree also implies that if $deg(F) < n - 1$ then G is not plateaued, and, in particular, not AB.

Note also that if F is APN, then G is differentially 6-uniform since the equation $D_aG(x) = b$ may have only $u_1, u_2, a + u_1$ and $a + u_2$ as solutions in addition to the two solutions of $D_aF(x) = b$. When $v_1 = v_2 = v$, G is in fact differentially 4-uniform because $D_aG(x) = b$ is equivalent to $D_aF(x) = b$ for $a = u_1 + u_2$, and for $a \neq u_1 + u_2$, $D_aG(u_1) = D_aG(u_2)$ implies

$$F(u_1) + F(u_2) = F(a + u_1) + F(a + u_2) + v + v = F(a + u_1) + F(a + u_2)$$

which contradicts the APN-ness of F.

Proposition 6 *Let F be an (n, n)-function satisfying $deg(F) < n$ and let $G(x) = F(x) + v(1_{u_1}(x) + 1_{u_2}(x))$. Then:*

(i) *the degree of G is $2^n - 2$ and $deg(G) = n - 1$ unless F contains a term cx^{2^n-2} for some $c \in \mathbb{F}_{2^n}^*$;*
(ii) $W_G(a, b) \in \{ W_F(a, b),\ W_F(a, b) \pm 4 \}$;
(ii) $N_G \in \{0, N_F \pm 2\}$;
(iv) *for $n \geq 4$, if F is AB then G is not AB and $N_G = 2^{n-1} - 2^{\frac{n-1}{2}} - 2$*
(v) *for $n \geq 5$, if F is plateaued then G is not plateaued;*
(vi) *for $n \geq 4$, if $deg(F) < n - 1$ then G is not plateaued, and, in particular, not AB.*

For investigating more complex conditions on the APN-ness of G, it can be useful to consider a simplified formulation wherein some of the values u_1, u_2, v_1, v_2 are fixed to e.g. 0 or 1. This can be done without loss of generality by constructing a function F' EA-equivalent to F and a function G' EA-equivalent to G as shown in the following two observations (recall that the property of a function being APN is invariant under EA-equivalence). Note that there are two cases depending on whether the values $F(u_1)$ and $F(u_2)$ are identical or not.

Observation 7 *Let F be an (n, n)-function and let G be defined as*

$$G(x) = F(x) + 1_{u_1}(x)v_1 + 1_{u_2}(x)v_2$$

for some $u_1, u_2, v_1, v_2 \in \mathbb{F}_{2^n}$ with $u_1 \neq u_2$ and $F(u_1) \neq F(u_2)$. Then there exists a function F' EA-equivalent to F satisfying $F'(0) = 0$ and $F'(1) = 1$, as well as $v_1', v_2' \in \mathbb{F}_{2^n}$ such that the function $G'(x)$ defined as

$$G'(x) = F'(x) + 1_0(x)v_1' + 1_1(x)v_2' \tag{7}$$

is EA-equivalent to G.

Proof Given F and G as above, the construction of G' is performed as follows: take $A_1(x) = \frac{x}{F(u_1)+F(u_2)}$, $A_2(x) = x(u_1 + u_2) + u_1$ and $A(x) = x + \frac{F(u_1)}{F(u_1)+F(u_2)}$; note that A_1 is indeed defined correctly due to $F(u_1) \neq F(u_2)$, that all three functions are affine and that A_1 and A_2 are bijective. Then the function $(A_1 \circ F \circ A_2 + A)$ is EA-equivalent to $G(x)$ and takes the form

$$[F(x(u_1 + u_2) + u_1) + (1 + (x(u_1 + u_2) + u_1 + u_1)^{2^n-1})v_1 + (1 + (x(u_1 + u_2) + u_1 + u_2)^{2^n-1})v_2 + F(u_1)] \cdot \frac{1}{F(u_1) + F(u_2)}. \tag{8}$$

Denoting $F'(x) = \frac{F(x(u_1+u_2)+u_1)+F(u_1)}{F(u_1)+F(u_2)}$, this becomes

$$F'(x) + (1+x^{2^n-1})\left(\frac{v_1}{F(u_1)+F(u_2)}\right) + (1+(x+1)^{2^n-1})\left(\frac{v_2}{F(u_1)+F(u_2)}\right). \quad (9)$$

If we now take $v_i' = \frac{v_i}{F(u_1)+F(u_2)}$ for $i \in \{1,2\}$, the above becomes

$$F'(x) + (1+x^{2^n-1})v_1' + (1+(x+1)^{2^n-1})v_2'. \quad (10)$$

It remains to observe that $F'(0) = 0$ and $F'(1) = 1$ which follows directly from the definition of F'. □

A similar approach can be used when $F(u_1) = F(u_2)$.

Observation 8 *Let F be an (n,n)-function and let G be defined as*

$$G(x) = F(x) + 1_{u_1}(x)v_1 + 1_{u_2}(x)v_2$$

for some $u_1, u_2, v_1, v_2 \in \mathbb{F}_{2^n}$ with $u_1 \neq u_2$ satisfying $F(u_1) = F(u_2)$, $v_i \neq 0$ for $i = 1, 2$. Then there exists a function F' EA-equivalent to F with $F'(0) = F'(1) = 0$, as well as some $v \in \mathbb{F}_{2^n}$ such that the function G' defined as

$$G'(x) = F'(x) + 1_0(x) + 1_1(x)v$$

is EA-equivalent to G.

Proof Given G as above, take $A_1(x) = \frac{x}{v_1}$, $A_2(x) = (u_1+u_2)x+u_1$ and $A(x) = x + \frac{F(u_1)}{v_1}$; note that all of these functions are well-defined linear permutations under the hypothesis. Then

$$\begin{gathered}(A_1 \circ G \circ A_2 + A)(x) = \frac{1}{v_1}[F(A_2(x)) + (1+(x(u_1+u_2))^{2^n-1})v_1 + \\ (1+((x+1)(u_1+u_2))^{2^n-1})v_2 + F(u_1)] = \\ (A_1 \circ F \circ A_2 + A)(x) + (1+x^{2^n-1}) + (1+(x+1)^{2^n-1})\frac{v_1}{v_2} = \\ (A_1 \circ F \circ A_2 + A)(x) + (1+x^{2^n-1}) + (1+(x+1)^{2^n-1})v\end{gathered} \quad (11)$$

for $v = v1/v2$; taking $F' = A_1 \circ F \circ A_2 + A$, it then suffices to verify that we have $F'(0) = F'(1) = 0$:

$$F'(0) = \frac{F(u_1)+F(u_1)}{v_1} = 0$$

and

$$F'(1) = \frac{F(u_2)+F(u_1)}{v_1} = 0$$

due to $F(u_1) = F(u_2)$. This completes the proof. □

5 Swapping two points

A significant simplification in the case of $F(u_1) \neq F(u_2)$ consists in examining the particular case of *swapping* the values of F at u_1 and u_2. This is a weaker construction, which makes it somewhat easier to obtain results. Furthermore, such a construction is of particular interest as it leaves the property of being a permutation invariant, i.e. if F is a permutation then G is a permutation too. In this case, the resulting function becomes (by substituting $v_1 = 1$ and $v_2 = 1$ into (7))

$$G(x) = F(x) + x^{2^n-1} + (x+1)^{2^n-1} \quad (12)$$

with $F(0) = 0$ and $F(1) = 1$.

Example 9 To see that swapping the values of a function F at two points is indeed weaker than the general case of changing two points, consider the function $F(x) = x^3 + (x^2 + x + 1)\mathrm{Tr}_1^n(x^3)$ over $\mathbb{F}_{2^4}$ of algebraic degree 3. All APN functions obtained by swapping two values of F have algebraic degree 2. On the other hand, taking e.g. $u_1 = \alpha^4$, $u_2 = \alpha^2$ and $v_1 = v_2 = 1$ (where α is a primitive element of $\mathbb{F}_{2^4}$) results in an APN function of algebraic degree 3.

In the following, we see how Lemmas 1 and 2 can be applied to the case of swapping two points. To facilitate the discussion, we define the following sets:

$$S = \{ (a, b) \in \mathbb{F}_{2^n} \ \mathrm{Tr}_1^n(a) = \mathrm{Tr}_1^n(b) = 1\}, \tag{13}$$

$$S' = \mathbb{F}_{2^n}^2 \setminus (S \cup \{(0, 0)\}). \tag{14}$$

The following lemma is known as the "two-tuple balance property" [11].

Lemma 10 *For $c \in \mathbb{F}_{2^n}$, we have*

$$\sum_{\substack{x\in\mathbb{F}_{2^n}\\ Tr(x)=1}} (-1)^{Tr(cx)} = \begin{cases} 2^{n-1} & c = 0 \\ -2^{n-1} & c = 1 \\ 0 & c \notin \{0, 1\}. \end{cases}$$

The following results easily follow by application of the above lemma.

Observation 11 *Let F be an (n, n)-function. Then*

$$\sum_{(a,b)\in S} W_F(a, b) = 2^{n-1} \sum_{\substack{b\in\mathbb{F}_{2^n}\\ Tr(b)=1}} (-1)^{Tr(bF(0))} - (-1)^{Tr(bF(1))}.$$

In particular, if $F(0) = 0$ and $F(1) = 1$, the above simplifies to

$$\sum_{(a,b)\in S} W_F(a, b) = 2^{2n-1}. \tag{15}$$

Proof From the definition, we have

$$\begin{aligned} \sum_{(a,b)\in S} W_F(a, b) = \sum_{(a,b)\in S} \sum_{x\in\mathbb{F}_{2^n}} \eta(\mathrm{Tr}(bF(x) + ax)) = \\ \sum_{x\in\mathbb{F}_{2^n}} \sum_{\mathrm{Tr}(b)=1} \eta_b(F(x)) \sum_{\mathrm{Tr}(a)=1} \eta_a(x) = \\ 2^{n-1} \cdot \left(\sum_{\mathrm{Tr}(b)=1} \eta_b(F(0)) - \sum_{\mathrm{Tr}(b)=1} \eta_b(F(1)) \right) = \\ 2^{n-1} \sum_{\mathrm{Tr}(b)=1} (\eta_b(F(0)) - \eta_b(F(1))) . \end{aligned} \tag{16}$$

The particular statement follows by substituting $F(0) = 0$ and $F(1) = 1$. □

Proposition 12 *Let F be an (n, n)-function. Then*

$$\sum_{(a,b)\in S} W_F^2(a, b) = 2^{3n-2} - 2^{2n-2}(\Delta_F(1, 0) - \Delta_F(1, 1)). \tag{17}$$

Proof From the definition we can easily obtain

$$\sum_{(a,b)\in S} W_F^2(a,b) = \sum_{x,y}\sum_{\mathrm{Tr}(b)=1} \eta_b(F(x)+F(y)) \sum_{\mathrm{Tr}(a)=1} \eta_a(x+y). \tag{18}$$

By Lemma 10, this is equivalent to

$$\begin{aligned}\sum_{(a,b)\in S} W_F^2(a,b) &= 2^{n-1}\sum_x \left(\sum_{\mathrm{Tr}(b)=1} \eta_b(F(x)+F(x)) - \sum_{\mathrm{Tr}(b)=1} \eta_b(F(x)+F(x+1))\right)\\ &= 2^{3n-2} - 2^{2n-2}(\Delta_F(1,0)-\Delta_F(1,1))\end{aligned} \tag{19}$$

which is what we wanted to show. □

Proposition 13 *Let F be an (n,n)-function satisfying $F(0)=0$ and $F(1)=1$. Then*

$$\sum_{a\in\mathbb{F}_{2^n}, Tr_1^n(a)=0} W_F(a,b) = 0, \tag{20}$$

$$\sum_{a\in\mathbb{F}_{2^n}, Tr_1^n(a)=1} W_F(a,b) = 2^n. \tag{21}$$

Proof Let $b\in\mathbb{F}_{2^n}$ satisfy $\mathrm{Tr}_1^n(b)=1$ and $G(x)=x^{2^n-1}+(x+1)^{2^n-1}+F(x)$. By Parseval's relation and by (6)

$$\begin{aligned}2^{2n} = \sum_{a\in\mathbb{F}_{2^n}} W_G^2(a,b) &= \sum_{\mathrm{Tr}_1^n(a)=0} W_F^2(a,b) + \sum_{\mathrm{Tr}_1^n(a)=1} (W_F(a,b)-4)^2\\ &= \sum_{a\in\mathbb{F}_{2^n}} W_F^2(a,b) - 8\sum_{\mathrm{Tr}_1^n(a)=1} W_F(a,b) + 2^{n+3}\\ &= 2^{2n} + 2^{n+3} - 8\sum_{\mathrm{Tr}_1^n(a)=1} W_F(a,b).\end{aligned}$$

Hence $\sum_{a\in\mathbb{F}_{2^n},\mathrm{Tr}_1^n(a)=1} W_F(a,b) = 2^n$. Besides, since $F(0)=0$ we have

$$\sum_{a\in\mathbb{F}_{2^n}} W_F(a,b) = 2^n$$

by the inverse Walsh transform (1) and thus $\sum_{a,\mathrm{Tr}_1^n(a)=0} W_F(a,b) = 0$. □

Proposition 14 *Let F be any function over $\mathbb{F}_{2^n}$ satisfying $F(0)=0$ and $F(1)=1$ and let G be defined as $G(x)=x^{2^n-1}+(x+1)^{2^n-1}+F(x)$. Then G is APN if and only if*

$$\begin{aligned}\sum_{(a,b)\in S}\left(W_F^3(a,b) - 6W_F^2(a,b)\right) =\\ 2^{-4}\sum_{a\in\mathbb{F}_{2^n}}\sum_{b\in\mathbb{F}_{2^n}^*} W_F^4(a,b) - 2^{4n-3} + 2^{3n-3} - 2^{2n+2},\end{aligned} \tag{22}$$

where S is defined as in (13). In particular, if F is APN, then G is APN if and only if

$$\sum_{(a,b)\in S}\left(W_F^3(a,b) - 6W_F^2(a,b)\right) = -2^{2n+2}. \tag{23}$$

Proof By (6) we get

$$\sum_{a\in\mathbb{F}_{2^n}}\sum_{b\in\mathbb{F}_{2^n}^*} W_G^4(a,b) = \sum_{(a,b)\in S'} W_F^4(a,b) + \sum_{(a,b)\in S} (W_F(a,b)-4)^4$$
$$= \sum_{a\in\mathbb{F}_{2^n}}\sum_{b\in\mathbb{F}_{2^n}^*} W_F^4(a,b) + \sum_{(a,b)\in S}\left(-16W_F^3(a,b)+96W_F^2(a,b)-256W_F(a,b)+256\right),$$

where S' is defined as in (14).

Hence, by (15) and Proposition 13

$$\sum_{(a,b)\in S}\left(W_F^3(a,b)-6W_F^2(a,b)\right) = 2^{-4}\sum_{a\in\mathbb{F}_{2^n}}\sum_{b\in\mathbb{F}_{2^n}^*} W_F^4(a,b) - 2^{-4}\sum_{a\in\mathbb{F}_{2^n}}\sum_{b\in\mathbb{F}_{2^n}^*} W_G^4(a,b) - 2^{2n+2},$$

and, by Lemma 1, G is APN if and only if (22) holds. If, in addition, F is APN then by Lemma 1, G is APN if and only if (15) is satisfied. □

We can now substitute the formula from Proposition 12 into Proposition 14 in order to obtain the following characterization.

Observation 15 *Let F be over $\mathbb{F}_{2^n}$ with $F(0)=0$ and $F(1)=1$, and let $G(x)=x^{2^n-1}+(x+1)^{2^n-1}+F(x)$. Then G is APN if and only if*

$$\sum_{(a,b)\in S} W_F^3(a,b)+3\cdot 2^{2n-1}(\Delta_F(1,0)-\Delta_F(1,1)) = 2^{-4}\sum_{a}\sum_{b\neq 0} W_F^4(a,b) + 13\cdot 2^{3n-3}-2^{4n-3}-2^{2n+2}. \quad (24)$$

If, in addition, F is APN, then G is APN if and only if

$$\sum_{(a,b)\in S} W_F^3(a,b) + 3\cdot 2^{2n-1}(\Delta_F(1,0)-\Delta_F(1,1)) = 3\cdot 2^{3n-1}-2^{2n+2}. \quad (25)$$

By means of mechanical computations similar to the ones in the proof of Proposition 12, we can obtain the following identity.

Proposition 16 *For any vectorial Boolean function F over $\mathbb{F}_{2^n}$ we have*

$$\sum_{(a,b)\in S} W_F^3(a,b) = 2^{2n-2}\sum_{y\in\mathbb{F}_{2^n}} (\Delta_F(y,F(y))-\Delta_F(y,F(y)+1)- \Delta_F(y+1,F(y))+\Delta_F(y+1,F(y)+1)). \quad (26)$$

In particular, if F is APN and satisfies $F(0)=0$ and $F(1)=1$, this becomes

$$\sum_{(a,b)\in S} W_F^3(a,b) = 3\cdot 2^{3n-1}-2^{2n}-2^{2n-2}\sum_{y\in\mathbb{F}_{2^n}} (\Delta_F(y,F(y)+1)+\Delta_F(y+1,F(y))). \quad (27)$$

Proof From the definition we have

$$\sum_{(a,b)\in S} W_F^3(a,b) = \sum_{x,y,z\in\mathbb{F}_{2^n}} \sum_{\substack{b\in\mathbb{F}_{2^n}\\ \mathrm{Tr}(b)=1}} \eta_b(F(x)+F(y)+F(z)) \sum_{\substack{a\in\mathbb{F}_{2^n}\\ \mathrm{Tr}(a)=1}} \eta_a(x+y+z)$$

$$= 2^{n-1} \sum_{x,y\in\mathbb{F}_{2^n}} \left(\sum_{\substack{b\in\mathbb{F}_{2^n}\\ \mathrm{Tr}(b)=1}} \eta_b(F(x)+F(y)+F(x+y)) \right.$$

$$\left. - \sum_{\substack{b\in\mathbb{F}_{2^n}\\ \mathrm{Tr}(b)=1}} \eta_b(F(x)+F(y)+F(x+y+1)) \right). \tag{28}$$

The main statement then follows by Lemma 10. The particular statement follows by observing that $\sum_{y\in\mathbb{F}_{2^n}} \Delta_F(y, F(y)) = \sum_{y\in\mathbb{F}_{2^n}} \Delta_F(y+1, F(y)+1) = 2^{n+1}+2^n-2$. □

Assuming that F is APN, we can combine (25) and (27) to obtain the following:

Theorem 17 *Let F be APN with $F(0) = 0$ and $F(1) = 1$. Then G is APN if and only if*

$$6\Delta_F(1,0) = \sum_{y\in\mathbb{F}_{2^n}} (\Delta_F(y, F(y)+1) + \Delta_F(y+1, F(y))). \tag{29}$$

This characterization allows us to show that a given G is not APN by computing a very weak lower bound on the number of pairs (x, y) satisfying $F(x)+F(x+y)+F(y) = 1$ or $F(x)+F(x+y+1)+F(y) = 0$. In Section 6 we derive a lower bound on the number of such pairs in the case of power functions. Experimental results show that the number of these solutions grows very quickly for APN functions as the dimension of the underlying field increases. Despite this, a theoretical lower bound appears difficult to prove in the general case; in Section 6.

Table 1 gives the values of both $\sum_y \Delta_F(y, F(y)+1)$ and $\sum_y \Delta_F(y+1, F(y))$ for all APN functions from the known switching classes over $\mathbb{F}_{2^n}$ with $5 \le n \le 8$ as given in [10]. The functions in the table are indexed using the notation from Tables 3, 5, 7 and 9 from [10]. Since the sum of the two values, $\sum_y \Delta_F(y, F(y)+1)$ and $\sum_y \Delta_F(y+1, F(y))$, is greater than 12 in all cases, we can immediately apply Theorem 17 to obtain the following result.

Since the sum of the two values $\sum_y \Delta_F(y, F(y)+1)$ and $\sum_y \Delta_F(y+1, F(y))$ for the functions in Table 1 in all cases is greater than 12, we can immediately apply Theorem 17 to get the following result.

Observation 18 *Let F be any APN function from those given in* [10]*, and let G be defined as*

$$G(x) = F(x) + x^{2^n-1} + (x+1)^{2^n-1}.$$

Then G is not APN.

Example 19 In some cases an explicit formula for the values $\sum_y \Delta_F(y, F(y)+1)$ and $\sum_y \Delta_F(y+1, F(y))$ can be found. Consider, for example, the function $F(x) = x^3$ over

Table 1 The values $S_1 = \sum_y \Delta_F(y, F(y)+1)$ and $S_2 = \sum_y \Delta_F(y+1, F(y))$ for all functions F from [10]

n	F	S_1	S_2	n	F	S_1	S_2
5	1.1	30	30		11.1	126	382
	1.2	30	30		12.1	126	382
	2.1	30	30		13.1	126	126
6	1.1	78	78		14.1	126	126
	1.2	78	78		14.2	126	382
	2.1	54	66		14.3	126	126
	2.2	54	78	8	1.1	222	222
	2.3	54	54		1.2	222	222
	2.4	54	54		1.3	222	222
	2.5	54	66		1.4	222	222
	2.6	42	54		1.5	270	222
	2.7	66	66		1.6	270	222
	2.8	78	54		1.7	270	222
	2.9	66	90		1.8	222	222
	2.10	54	42		1.9	270	222
	2.11	66	66		1.10	222	222
	2.12	54	66		1.11	270	222
7	1.1	126	126		1.12	222	222
	1.2	126	382		1.13	270	222
	2.1	126	126		1.14	270	270
	2.2	126	382		1.15	222	222
	3.1	126	126		1.16	222	222
	4.1	126	126		1.17	222	222
	5.1	126	126		2.1	270	222
	6.1	126	126		3.1	270	222
	7.1	126	126		4.1	318	766
	8.1	126	126		5.1	318	318
	9.1	126	382		6.1	318	318
	10.1	126	382		7.1	222	222
	10.2	126	126				

$\mathbb{F}_{2^n}$. The sum $\sum_y \Delta_F(y, F(y)+1)$ is equal to the number of pairs $(x, y) \in \mathbb{F}_{2^n}^2$ satisfying $x^3 + (x+y)^3 + y^3 = 1$, which easily simplifies to

$$x^2y + xy^2 = 1. \tag{30}$$

Note that if $y = 0$, no pair $(x, 0)$ can satisfy (30). Consider some fixed nonzero $y \in \mathbb{F}_{2^n}^*$. Dividing both sides of (30) by the cube of its inverse yields

$$\left(\frac{x}{y}\right)^2 + \left(\frac{x}{y}\right) = \frac{1}{y^3}. \tag{31}$$

For this fixed y, define the function $g_y(x) = \left(\frac{x}{y}\right)^2 + \left(\frac{x}{y}\right)$. Observe that $g_y(x_1) = g_y(x_2)$ implies

$$\left(\frac{x_1 + x_2}{y}\right)^2 = \left(\frac{x_1 + x_2}{y}\right)$$

which is certainly satisfied when $x_1 = x_2$. If $x_1 \neq x_2$, however, we can divide both sides by $(x_1 + x_2)/y$ in order to obtain

$$x_1 + x_2 = y. \tag{32}$$

Thus, $g_y(x)$ is a 2-to-1 function on $\mathbb{F}_{2^n}$ for any $y \in \mathbb{F}_{2^n}^*$. Taking into account that $\mathrm{Tr}_n(g_y(x)) = 0$ for any $x \in \mathbb{F}_{2^n}$, we can see that $\{g_y(x) : x \in \mathbb{F}_{2^n}\} = \{x \in \mathbb{F}_{2^n} : \mathrm{Tr}_n(x) = 0\}$. This means that for any fixed $y \in \mathbb{F}_{2^n}^*$, (30) has two solutions if $\mathrm{Tr}_n(y^{-3}) = 0$, and no

solutions otherwise. The value $\sum_y \Delta_F(y, F(y)+1)$ is then equal to twice the number of nonzero elements $y \in \mathbb{F}_{2^n}$ with $\mathrm{Tr}_n(y^3) = 0$, i.e.

$$\sum_y \Delta_F(y, F(y)+1) = 2(wt(f) - 1) \tag{33}$$

where f is the function $f(a) = \mathrm{Tr}_n(a^3)$, and $wt(f) = |\{x \in \mathbb{F}_{2^n} : f(x) \neq 0\}|$ is the Hamming weight of f.

The weight of any function of the form $x \mapsto \mathrm{Tr}_n(\lambda x^3)$, where $\lambda \in \mathbb{F}_{2^n}$ is a constant, is known from [7]. Adapting this slightly more general result to our particular case and substituting in (33), we have

$$\sum_y \Delta_F(y, F(y)+1) = \begin{cases} 2(2^{n-1} - 1) & n \text{ is odd} \\ 2(2^{n-1} + 2^{n/2} - 1) & n \text{ is even and } n/2 \text{ is odd} \\ 2(2^{n-1} - 2^{n/2} - 1) & n \text{ is even and } n/2 \text{ is even.} \end{cases} \tag{34}$$

This formula then allows us to compute the values of $\sum_y \Delta_F(y, F(y)+1)$ for $F(x) = x^3$ for any dimension n. Table 2 lists the values obtained for all dimensions up to 20. Since these values are at least 30 for $n \geq 5$, we can make the following observation.

Observation 20 *Let $F(x) = x^3$ over $\mathbb{F}_{2^n}$ and let G be defined as*

$$G(x) = F(x) + x^{2^n-1} + (x+1)^{2^n-1},$$

i.e. by swapping the values of F at 0 and 1. Then G is not APN unless $n \leq 4$.

We have experimentally verified that all power APN functions over fields $\mathbb{F}_{2^n}$ of dimension $1 \leq n \leq 10$ have the same exact values for $\sum_y \Delta_F(y, F(y)+1)$ as those in Table 2. We thus formulate the following conjecture.

Conjecture 21 For any power APN function F over $\mathbb{F}_{2^n}$, the two values $\sum_y \Delta_F(y, F(y)+1)$ and $\sum_y \Delta_F(y+1, F(y))$ are equal to $\sum_y \Delta_{x^3}(y, y^3+1)$ and are given by (34).

Furthermore, the results in Table 1 suggest that even in the case of a general APN function, the values do not differ too much from that of the power APN function x^3 over the

Table 2 Formula (34) applied to $F(x) = x^3$ for dimensions $1 \leq n \leq 20$

n	$\sum_y \Delta_F(y, F(y)+1)$	n	$\sum_y \Delta_F(y, F(y)+1)$
1	0	11	2046
2	6	12	3966
3	6	13	8190
4	6	14	16638
5	30	15	32766
6	78	16	65022
7	126	17	131070
8	222	18	263166
9	510	19	524286
10	1086	20	1046526

same field. Seeing as how the latter form a sequence that monotonically increases with the dimension n, we formulate the following conjecture.

Conjecture 22 Let F be any APN function over $\mathbb{F}_{2^n}$ with $n \geq 5$. Then no function G obtained by swapping two points of F can be APN.

Note also that a similar computation can be performed in order to obtain an explicit formula for the value $\sum_y \Delta_F(y+1, F(y))$. In the case of $F(x) = x^3$, we obtain the same formula as for $\sum_y \Delta_F(y, F(y)+1)$ so that in this case the two values are always the same. Based on our experimental results, this is also true for all power APN functions over $\mathbb{F}_{2^n}$ with $1 \leq n \leq 10$. As can be seen from Table 1, however, the two values may differ in the general case.

6 Power functions

In this section we apply the conditions from Section 5 to show that for $n \geq 5$ no APN function can be obtained by swapping the values $F(0)$ and $F(1)$ of a power function. A well-known observation is that a power function F is APN if and only if its derivative D_1F is a 2-to-1 function.

Recall that we assume $u_1 = 0$, $u_2 = 1$ and $v_1 = v_2 = v = 1$, as well as $F(0) = 0$ and $F(1) = 1$. Then condition (iii) from Proposition 4 becomes: for $a \neq 1$, the equations $D_aF(x) + F(a) = 1$ and $D_aF(x) + F(a+1) = 0$ have no solution $x \notin \{0, 1, a, a+1\}$. We will show that when F is a power function of sufficiently high dimension, a pair (a, x) violating this condition always exists.

Since $F(x) = x^l$ for some positive integer l, the equation e.g. $D_1F(x) + F(a) = 1$ can be rewritten as

$$x^l + (a+x)^l + a^l = 1.$$

We divide the analysis into several cases depending on the parity of the dimension n. We first show that any field of odd dimension contains at least one pair (a, x) satisfying $D_1F(x) + F(a) = 1$.

Observation 23 *Let $\mathbb{F}_{2^n}$ be a finite field with $n > 1$ odd, and let $F(x) = x^l$ be a power APN function over $\mathbb{F}_{2^n}$. Then there exists a pair $(a, x) \in \mathbb{F}_{2^n}^2$ satisfying $a^l + x^l + (a+x)^l = 1$.*

Proof Note that the equation $a^l + x^l + (a+x)^l = 1$ can be more succinctly written as $D_a^0F(x) = 1$. Since x^l is a permutation, it suffices to find a pair (a, x) such that $D_a^0F(x) \neq 0$. Since F is APN, the shifted derivative D_a^0 is a 2-to-1 function for any fixed $a \neq 0$, and maps all elements of $\mathbb{F}_{2^n}$ to 0 for $a = 0$. Thus, at most $2^n + 2(2^n - 1)$ pairs (a, x) can satisfy $D_a^0F(x) = 0$. This quantity is strictly less than 2^{2n} for $n > 1$, so that at least one pair (a, x) must exist satisfying $D_a^0F(x) = c$ for some $c \neq 0$. It now suffices to divide both sides by c^{-1}. □

It is easy to show that such a pair (a, x) exists for any APN power function over $\mathbb{F}_{2^2}$ as well.

Observation 24 *Let $F(x) = x^l$ be APN over $\mathbb{F}_{2^2}$. Then there is a pair $(a, x) \in \mathbb{F}_{2^n}^2$ such that $x^l + (a+x)^l + a^l = 1$.*

Proof Let α be a primitive element of $\mathbb{F}_{2^2}$. Since $F(x) = x^l$ is APN over a field of even dimension n, we must have $\gcd(l, n) = 3$. Then l is a multiple of three, say $l = 3l'$ for some positive integer l'. Now

$$\alpha^l + (\alpha + 1)^l + 1^l = \alpha^{3l'} + (\alpha + 1)^{3l'} + 1 = 1 + 1 + 1 = 1$$

since $\alpha^3 = (\alpha + 1)^3 = 1$. □

A simple but important observation is that for any two pairs (a_1, x_1) and (a_2, x_2) satisfying $D^0_{a_1} F(x_1) = D^0_{a_2} F(x_2) = 1$, the sets $\{x_1, a_1 + x_1, a_1\}$ and $\{x_2, a_2 + x_2, a_2\}$ either coincide or are disjoint.

Observation 25 *Let F be APN over $\mathbb{F}_{2^n}$ with $F(0) = 0$ and $F(1) = 1$. Let $A_1 = \{a_1, b_1, a_1 + b_1\}$ and $A_2 = \{a_2, b_2, a_2 + b_2\}$ be two sets of elements defining solutions to the equation*

$$F(a_i) + F(a_i + b_i) + F(b_i) = 1 \tag{35}$$

for $i \in \{1, 2\}$. Then A_1 and A_2 are either identical or disjoint.

We can now apply the above observations in order to show that swapping two points in a power function cannot produce an APN function except over $\mathbb{F}_2$, $\mathbb{F}_{2^2}$ and $\mathbb{F}_{2^4}$.

Proposition 26 *Let $F(x) = x^l$ be a power function over $\mathbb{F}_{2^n}$ and $G(x) = x^l + x^{2^n-1} + (x + 1)^{2^n-1}$. Then G cannot be APN for any dimension n other than 1, 2 and 4.*

Proof Let us assume that G is APN. First, observe that the derivatives of F and G in direction 1 coincide since

$$D_1 G(x) = F(x) + x^{2^n-1} + (x + 1)^{2^n-1} + F(x + 1) + (x + 1)^{2^n-1} + x^{2^n-1} = D_1 F(x).$$

Since $F(x) = x^l$ is APN if and only if $D_1 F$ is 2-to-1, if G is APN then F must necessarily be APN as well.

We now examine several cases.

If n is a composite number and p and q are two distinct primes dividing n, then F is APN over $\mathbb{F}_{2^p}$ and $\mathbb{F}_{2^q}$ as well. According to Observations 23 and 24, there exist pairs $(a_1, x_1) \in \mathbb{F}^2_{2^p}$ and $(a_2, x_2) \in \mathbb{F}^2_{2^q}$ such that $D_{a_i} F(x) + F(a_i) = 1$ for $i \in \{1, 2\}$. Since p and q are coprime and therefore $\mathbb{F}_{2^p} \cap \mathbb{F}_{2^q} = \mathbb{F}_2$, the sets $\{a_1, x_1, a_1 + x_1\}$ and $\{a_2, x_2, a_2 + x_2\}$ are disjoint by Observation 25.

Note that if some pair (a, x) solves $D_a F(x) + F(a) = 1$, then so do the pairs (x, a), $(a+x, a)$, $(a+x, x)$, $(a, a+x)$ and $(x, a+x)$. Furthermore, all of these pairs are distinct as can be easily seen by contradiction. Hence the two pairs (a_1, x_1) and (a_2, x_2) above actually yield 12 distinct pairs (a, x) solving $D_a F(x) + F(a) = 1$. This means that the right-hand side of (29) from Theorem 17 is at least 24, while the left-hand side is at most 12 due to F being APN. Then according to Theorem 17, G is not APN.

If n is odd, then by Observation 23, there exists a pair (a, x) solving $D_a F(x) + F(a) = 1$. However, due to n being odd, F is a permutation over $\mathbb{F}_{2^n}$ and thus $\Delta_F(1, 0) = 0$. Thus, the left-hand side of (29) is equal to zero while its right-hand side is positive. Consequently, Theorem 17 implies that G is not APN.

Finally, suppose that n is a power of two, i.e. $n = 2^k$ for some $k > 2$. Since G is APN over $\mathbb{F}_{2^{2^k}}$ it must be APN over $\mathbb{F}_{2^{2^{k'}}}$ for all $k' < k$. In particular, G must be APN over $\mathbb{F}_{2^8}$. However, no power APN function over $\mathbb{F}_{2^8}$ produces an APN function after a two-point

swap, which can be verified by an exhaustive search. This means that G cannot be APN in this case either. □

To complete the discussion, it is worth mentioning that $G(x)$ can indeed be APN for $n \in \{1, 2, 4\}$. This can easily be verified by using e.g. Theorem 17 or the definition of APN functions.

Observation 27 *Over the fields $\mathbb{F}_{2^{2^i}}$ for $i \in \{0, 1, 2\}$, the function $G(x) = x^l + x^{2^n-1} + (x+1)^{2^n-1}$ is APN if and only if $F(x) = x^l$ is APN, where $n = 2^i$.*

7 Equivalence relations

In this section we make some observations regarding the EA- and CCZ-equivalence classes of the functions F and G from the two-point construction.

We begin by observing that the possibility of obtaining one EA-equivalence class from another does not depend on the concrete choice of representatives.

Proposition 28 *Let $F : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$ and G be defined as $G(x) = F(x) + v_1(x+u_1)^{2^n-1} + v_2(x+u_2)^{2^n-1}$ for some $u_1, u_2 \in \mathbb{F}_{2^n}$ and $v_1, v_2 \in \mathbb{F}_{2^n}^*$, and let $G' : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$ be EA-equivalent to G. Then there exist elements $u_1, u_2' \in \mathbb{F}_{2^n}$ and $v_1', v_2' \in \mathbb{F}_{2^n}^*$ and a function $F' : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$ EA-equivalent to F such that*

$$G'(x) = F'(x) + v_1'(x+u_1')^{2^n-1} + v_2'(x+u_2')^{2^n-1}.$$

Proof Suppose that G' is EA-equivalent to G via $G' = A_1 \circ G \circ A_2 + A$ where A_1, A_2, A are affine functions over $\mathbb{F}_{2^n}$ and A_1, A_2 are permutations. Assume without loss of generality that A_1 is linear. We can write

$$\begin{aligned} G'(x) = A_1(F(A_2(x))) + A_1(v_1(A_2(x)+u_1)^{2^n-1}) + \\ A_1(v_2(A_2(x)+u_2)^{2^n-1}) + A(x) = (A_1 \circ F \circ A_2 + A)(x) + \\ A_1(v_1)(x + A_2^{-1}(u_1))^{2^n-1} + A_1(v_2)(x + A_2^{-1}(u_2))^{2^n-1}. \end{aligned} \quad (36)$$

If we now take $F' = A_1 \circ F \circ A_2 + A$, $u_i' = A_1(v_i)$ and $v_i' = A_2^{-1}(v_i)$ for $i \in \{1, 2\}$, we have $G'(x) = F'(x) + v_1'(x+u_1')^{2^n-1} + v_2'(x+u_2)^{2^n-1}$ with F' clearly being EA-equivalent to F and with $v_1', v_2' \neq 0$ due to A_2 being a permutation. □

The following proposition is obtained by a similar argument.

Proposition 29 *Let F and F' be EA-equivalent functions over $\mathbb{F}_{2^n}$ and let G be defined by $G(x) = F(x) + v_1(x+u_1)^{2^n-1} + v_2(x+u_2)^{2^n-1}$ for some $u_1, u_2 \in \mathbb{F}_{2^n}$ and $v_1, v_2 \in \mathbb{F}_{2^n}^*$. Then there is a function G' EA-equivalent to G and some elements $u_1', u_2' \in \mathbb{F}_{2^n}$ and $v_1', v_2' \in \mathbb{F}_{2^n}^*$ such that*

$$G'(x) = F'(x) + v_1'(x+u_1')^{2^n-1} + v_2'(x+u_2')^{2^n-1}.$$

We now turn to the more general notion of CCZ-equivalence.

Proposition 30 *Let F be a function over $\mathbb{F}_{2^n}$ and let G be defined over $\mathbb{F}_{2^n}$ as*

$$G(x) = F(x) + v_1(x+u_1)^{2^n-1} + v_2(x+u_2)^{2^n-1}$$

for some $u_1, u_2 \in \mathbb{F}_{2^n}$ and $v_1, v_2 \in \mathbb{F}_{2^n}^*$, and suppose that G' is CCZ-equivalent to G. Then there exists a function F' CCZ-equivalent to F and elements $u_1', u_2', v_1', v_2' \in \mathbb{F}_{2^n}$ such that

$$G'(x) = F'(x) + v_1'(x + u_1')^{2^n-1} + v_2'(x + u_2')^{2^n-1}.$$

Proof Suppose that G' is CCZ-equivalent to G via the affine permutation $\mathcal{L}(x, y) = (L_1(x, y), L_2(x, y))$ of $\mathbb{F}_{2^n}^2$. Writing $G_i(x) = L_i(x, G(x))$ for $i \in \{1, 2\}$ and $A_1(x, y) = L_1(x) + L_2(y) + a$ and $A_2(x, y) = L_3(x) + L_4(y) + b$ for L_1, L_2, L_3, L_4 linear and $a, b \in \mathbb{F}_{2^n}$, we have

$$\begin{aligned} G'(x) = L_3(G_1^{-1}(x)) + L_4(G(G_1^{-1}(x))) + b = L_3 \circ G_1^{-1}(x) + L_4 \circ F \circ G_1^{-1}(x) + \\ L_4((G_1^{-1}(x) + u_1)^{2^n-1} v_1) + L_4((G_1^{-1}(x) + u_2)^{2^n-1} v_2) + b = F'(x) + \\ L_4((G_1^{-1}(x) + u_1)^{2^n-1} v_1 + (G_1^{-1}(x) + u_2)^{2^n-1} v_2) = \\ F'(x) + L_4(v_1)(x + G_1(u_1))^{2^n-1} + L_4(v_2)(x + G_1(u_2)) \end{aligned}$$

where we define $F'(x) = F_2 \circ G_1^{-1}(x)$ with $F_2(x) = L_3(x) + L_4(F(x)) + b$. Now F' is clearly CCZ-equivalent to F by the construction, and it suffices to take $u_i' = G_1(u_i)$ and $v_i' = L_4(v_i)$ for $i \in \{1, 2\}$. □

Note that in the above proposition, we do not guarantee that v_1' and v_2' will be nonzero which means that the functions F' and G' may be at distance less than two. This raises the question of whether the minimum distance between functions belonging to distinct CCZ-equivalence classes depends on the concrete choice of representatives.

8 Conclusion

We investigated a construction that involves changing the value of a given function F over $\mathbb{F}_{2^n}$ at precisely two points of $\mathbb{F}_{2^n}$ and, in particular, concentrated on the possibility of obtaining an APN function in this manner when F is APN. We characterized the APN-ness of the resulting function G in terms of the derivatives and in terms of the Walsh coefficients of F. We observed that changing two points in a plateaued function F cannot produce a plateaued function G, and neither can this happen for a function F with $deg(F) < n - 1$ (unless $n < 5$). Furthermore, no two AB functions at distance two may exist for $n \geq 4$.

In addition, we observed that if $deg(F) = 2$ and F is not APN, then any function G obtained by changing two points of F cannot be APN either.

We also examined the less general but more tractable problem of swapping two points in a given function. In particular, we showed that if F is APN and satisfies $F(0) = 0$ and $F(1) = 1$, then G is APN if and only if

$$6\Delta_F(1, 0) = \sum_{y \in \mathbb{F}_{2^n}} \Delta_F(y, F(y) + 1) + \Delta_F(y + 1, F(y)).$$

We applied this characterization in order to show that an APN function can never be obtained by swapping two values in any power function over $\mathbb{F}_{2^n}$ for $n > 4$. We also derived an explicit formula for $\sum_y \Delta_F(y, F(y) + 1)$ in the case of $F(x) = x^3$. Furthermore, we experimentally computed the values $\Delta_F(y, F(y)+1)$ and $\Delta_F(y+1, F(y))$ all known APN functions from [10], concluding that no APN function can be obtained by a two-point swap from those functions either.

Despite the fact that the results in Tables 1 and 2 suggest that the values of $\Delta_F(y, F(y)+1)$ and $\Delta_F(y+1, F(y))$ grow quite quickly with the dimension n of the underlying field, it seems difficult to find a lower bound on those values in the general case. Further results in this direction should provide additional insight into the possibility of obtaining an APN functions by a two-point swap.

The general case of arbitrarily changing two points in a given function also leaves room for further investigation; in particular, an analogue to Theorem 17 is highly desirable.

Finally, as mentioned above, we mostly concentrated on the case when the initial function F is itself APN. The possibility of obtaining an APN function G from a function F belonging to some other class of functions is another potential direction for future work.

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